

Representations of Dynkin Quivers

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1. Introduction

2. Representations of Quivers

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1. Introduction

Quiver representation is an important concept in algebraic representation theory. It provides numerous instances for algebraic representation theory and has significant applications in areas such as bundle algebras and algebraic geometry.

This program aims at investigating the basic theory of quiver representation theory, and calculate the minimal projective resolution of the indecomposable modules of finite-dimensional representation of all type A_n Dynkin quivers and a special class of type D_n Dynkin quivers.

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2.1 Path Algebra

Definition 2.1.1 (Quiver) A quiver $Q = (Q_0, Q_1, s, t : Q_1 \rightarrow Q_0)$ is given by a set Q_0 of vertices and a set Q_1 of arrows. An arrow ρ starts at the vertex $s(\rho)$ and terminates at the vertex $t(\rho)$.

Basically speaking, a “quiver” is a directed graph. This concept is proposed by Peter Gabriel in his paper in 1972 that laid the foundation for quiver representation theory, since the term “graph” already includes too many related concepts.

2.1 Path Algebra

Definition 2.1.2 (Path) A (non-trivial) path in Q is a sequence $\rho_m \dots \rho_1$ ($m \geq 1$) of arrows which satisfies $t(\rho_i) = s(\rho_{i+1})$ for $1 \leq i < m$.



Definition 2.1.3 (Path Algebra) The path algebra kQ is the k -algebra with basis all the possible paths in Q , and with the products of two paths x, y given by

$$xy := \begin{cases} \text{Composition of paths} & \text{if } t(y) = s(x) \\ 0 & \text{otherwise} \end{cases}$$

2.2 Representations

We now define a category $\text{Rep}(Q)$ of representations of Q .

Definition 2.2.1 (Representation of Quiver) A representation V of Q is given by a k -vector space V_i for each $i \in Q_0$ and a linear map $V_\rho : V_{s(\rho)} \rightarrow V_{t(\rho)}$ for each $\rho \in Q_1$.

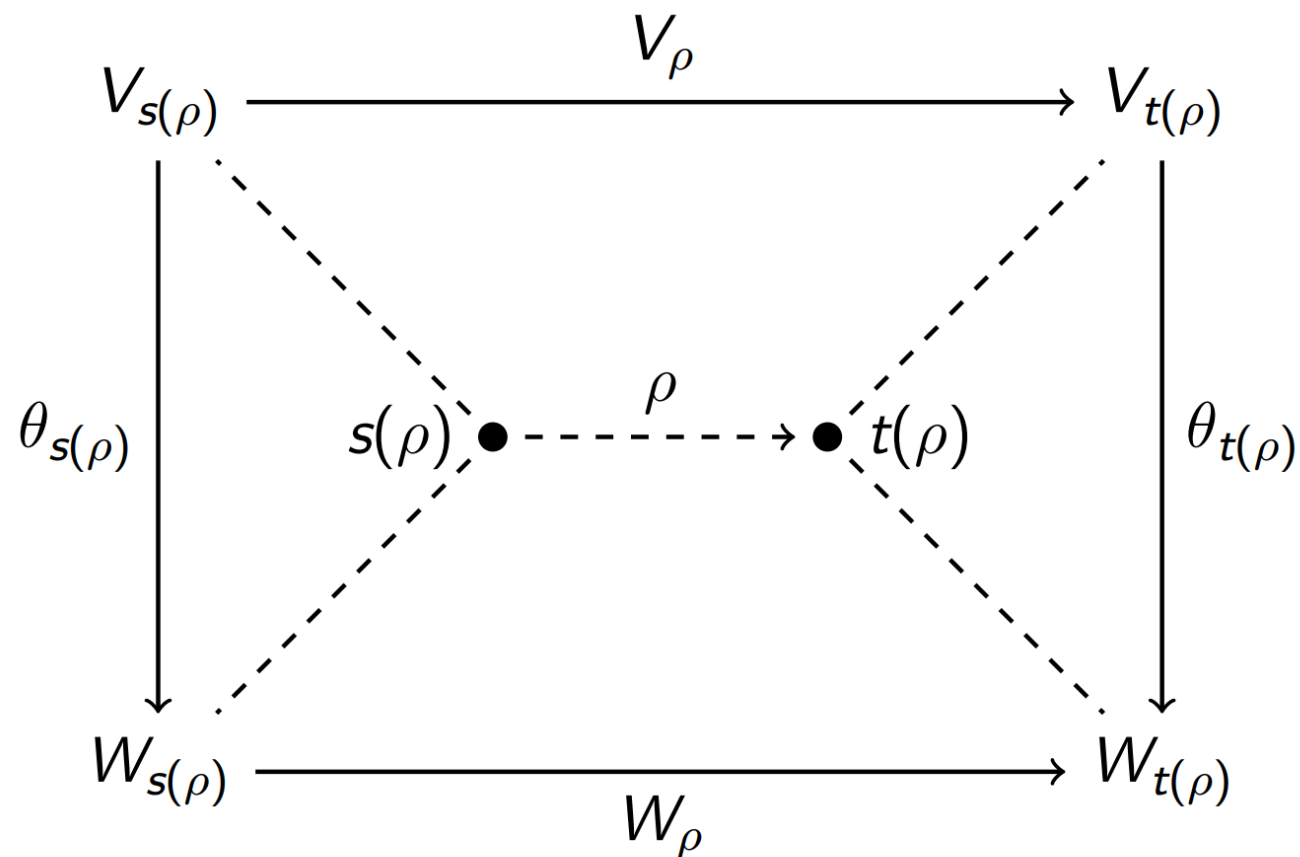
A morphism $\theta : V \rightarrow W$ is given by linear maps $\theta_i : V_i \rightarrow W_i$ for each $i \in Q_0$ satisfying

$$W_\rho \circ \theta_{s(\rho)} = \theta_{t(\rho)} \circ V_\rho$$

for each $\rho \in Q_1$.

2.2 Representations

We can summarize the definitions above through the following diagram.



2.2 Representations

Theorem 2.2.2 Given a quiver $Q = (Q_0, Q_1, s, t)$, then $\text{Rep}(Q) \cong \text{Mod}_{kQ}$.

Proof.

- $F : \text{Rep}(Q) \rightarrow \text{Mod}_{kQ}$

For $V \in \text{Rep}(Q)$, define $F(V) = \bigoplus_{i \in Q_0} V_i$. For path $\rho_m \cdots \rho_1$, define it acts as

$$\rho_m \cdots \rho_1 v = \varepsilon_{t(\rho_m)} \circ V_{\rho_m} \circ \cdots \circ V_{\rho_1} \circ \pi_{s(\rho_1)} v$$

- $G : \text{Mod}_{kQ} \rightarrow \text{Rep}(Q)$

For $X \in \text{Mod}_{kQ}$, define $G(X)_i = e_i X$, and define $G(X)_\rho : G(X)_{s(\rho)} \rightarrow G(X)_{t(\rho)}$ as $x \mapsto \rho x$. □

2.3 Gabriel's theorem

A_n : $\bullet \text{---} \bullet \text{---} \bullet \dots \bullet \text{---} \bullet$

D_n : $\begin{array}{c} \bullet \\ | \\ \bullet \text{---} \bullet \text{---} \bullet \dots \bullet \text{---} \bullet \end{array}$

E_6 : $\begin{array}{c} \bullet \\ | \\ \bullet \text{---} \bullet \text{---} \bullet \text{---} \bullet \text{---} \bullet \end{array}$

E_7 : $\begin{array}{c} \bullet \\ | \\ \bullet \text{---} \bullet \text{---} \bullet \text{---} \bullet \text{---} \bullet \text{---} \bullet \end{array}$

E_8 : $\begin{array}{c} \bullet \\ | \\ \bullet \text{---} \bullet \text{---} \bullet \text{---} \bullet \text{---} \bullet \text{---} \bullet \text{---} \bullet \end{array}$

2.3 Gabriel's theorem

Theorem 2.3.1 (Gabriel's Theorem)

- A (connected) quiver is of finite type if and only if its underlying graph (when the directions of the arrows are ignored) is one of the ADE Dynkin diagrams: A_n, D_n, E_6, E_7, E_8 .
- The indecomposable representations are in a one-to-one correspondence with the positive roots of the root system of the Dynkin diagram.

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3.1 Standard Resolution

There is a standard resolution of quiver [1].

Theorem 3.1.1 Let $A = kQ$ and M be a left A -module. There exists an exact sequence

$$0 \rightarrow \bigoplus_{\rho \in Q_1} Ae_{t(\rho)} \otimes_k e_{s(\rho)} M \xrightarrow{f} \bigoplus_{i=1}^n Ae_i \otimes_k e_i M \xrightarrow{g} M \rightarrow 0$$

where $g(a \otimes m) = am$ if $a \in Ae_i$, $m \in e_i M$,

$f(a \otimes m) = a\rho \otimes m - a \otimes \rho m$ if $a \in Ae_{t(\rho)}$, $m \in e_{s(\rho)} M$.

$$Ae_i \triangleq P(i) = (\dim(e_1 Ae_i), \dim(e_2 Ae_i), \dots, \dim(e_n Ae_i))$$

3.2 Minimal Projective Resolution

Definition 3.2.1 (Projective Cover) Let M be a module and P be a projective module. A surjection $\varphi : P \rightarrow M$ is called a projective cover if for all submodule P' of P , $\varphi|_{P'}$ is not surjective.

Definition 3.2.2 (Minimal Projective Resolution) A projective resolution of a module M

$$\cdots \rightarrow P_2 \xrightarrow{d_1} P_1 \xrightarrow{d_0} P_0 \xrightarrow{\pi} M \rightarrow 0$$

is called minimal projective resolution if both π and all of $d_n^* : P_{n+1} \rightarrow \ker(d_{n-1})$ are projective covers.

3.2 Minimal Projective Resolution

Definition 3.2.4 (Projective Cover) Let M be a module and P be a projective module. A surjection $\varphi : P \rightarrow M$ is called a projective cover if for all submodule P' of P , $\varphi|_{P'}$ is not surjective.

Definition 3.2.5 (Minimal Projective Resolution) A projective resolution of a module M

$$\cdots \rightarrow P_2 \xrightarrow{d_1} P_1 \xrightarrow{d_0} P_0 \xrightarrow{\pi} M \rightarrow 0$$

is called minimal projective resolution if both π and all of $d_n^* : P_{n+1} \rightarrow \ker(d_{n-1})$ are projective covers.

Proposition 3.2.6 If $\varphi_1 : P_1 \rightarrow M_1$ and $\varphi_2 : P_2 \rightarrow M_2$ are two projective covers, then so is $\varphi_1 \oplus \varphi_2 : P_1 \oplus P_2 \rightarrow M_1 \oplus M_2$.

3.2 Minimal Projective Resolution

Standard resolution isn't necessary to be minimal.

Example. Consider the quiver

$$1 \longrightarrow 2 \longrightarrow 3$$

and one of its representation M

$$k \xrightarrow{\text{id}} k \longrightarrow 0$$

The standard resolution of M is

$$0 \rightarrow P(2) \oplus P(3) \xrightarrow{f} P(1) \oplus P(2) \xrightarrow{g} M \rightarrow 0$$

However, $P(1)$ is a submodule of $P(1) \oplus P(2)$ that makes $g|_{P(1)}$ surjective, so this resolution is not minimal.

In fact, the minimal projective resolution of M is

$$0 \rightarrow P(3) \xrightarrow{f} P(1) \xrightarrow{g} M \rightarrow 0$$

3.3 Minimal Projective Resolution for Type A_n

Positive roots of A_n are as follow:

$$0 - 0 - \dots - 0 - 1 - 1 - \dots - 1 - 0 - \dots - 0$$

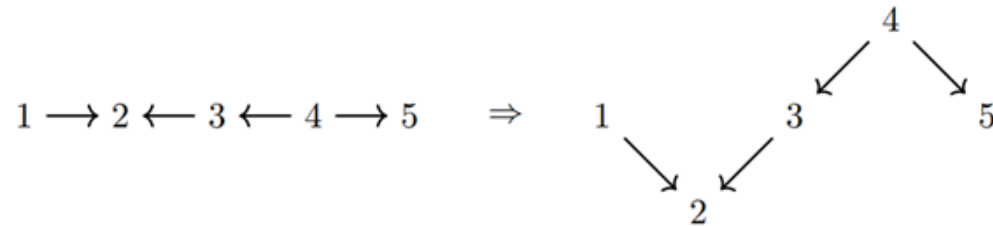
Therefore the indecomposable representations of a quiver with underlying graph A_n are of the form

$$U_{ij} = (\overbrace{0, \dots, 0}^{i-1}, \overbrace{1, \dots, 1}^{j-i+1}, \overbrace{0, \dots, 0}^{n-j})$$

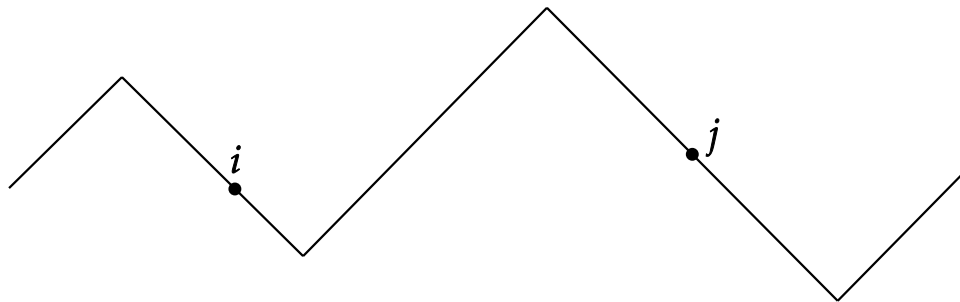
and it can be shown that all morphisms between two k 's should be identity.

3.3 Minimal Projective Resolution for Type A_n

We can draw the quiver as a zigzag partially ordered set according to the arrows:



Denote $T(i, j)$ as the set of all source point in (i, j) and $B(i, j)$ as the set of all sink point in (i, j) .



3.3 Minimal Projective Resolution for Type A_n

Theorem 3.3.1 For a indecomposable representation U_{ij} of a quiver with underlying graph A_n , if the arrows between $(i, i + 1)$ and $(j - 1, j)$ are as

- $i \leftarrow i + 1, j - 1 \rightarrow j$, then the minimal projective resolution of U_{ij} is as

$$0 \rightarrow P(i - 1) \oplus P(j + 1) \oplus \bigoplus_{m \in B(i, j)} P(m) \rightarrow \bigoplus_{m \in T(i, j)} P(m) \rightarrow U_{ij} \rightarrow 0$$

- $i \rightarrow i + 1, j - 1 \rightarrow j$, then the minimal projective resolution of U_{ij} is as

$$0 \rightarrow P(j + 1) \oplus \bigoplus_{m \in B(i, j)} P(m) \rightarrow P(i) \oplus \bigoplus_{m \in T(i, j)} P(m) \rightarrow U_{ij} \rightarrow 0$$

- $i \leftarrow i + 1, j - 1 \leftarrow j$, then the minimal projective resolution of U_{ij} is as

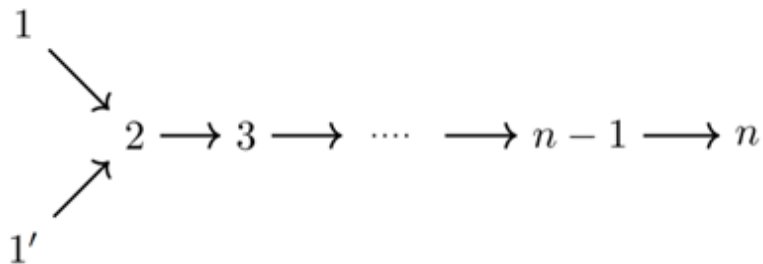
$$0 \rightarrow P(i - 1) \oplus \bigoplus_{m \in B(i, j)} P(m) \rightarrow P(j) \oplus \bigoplus_{m \in T(i, j)} P(m) \rightarrow U_{ij} \rightarrow 0$$

- $i \rightarrow i + 1, j - 1 \leftarrow j$, then the minimal projective resolution of U_{ij} is as

$$0 \rightarrow \bigoplus_{m \in B(i, j)} P(m) \rightarrow P(i) \oplus P(j) \oplus \bigoplus_{m \in T(i, j)} P(m) \rightarrow U_{ij} \rightarrow 0$$

3.4 Minimal Projective Resolution for Type D_n

We now consider a special class of quivers of type D_{n+1} :



3.4 Minimal Projective Resolution for Type D_n

As shown in [2], all of its indecomposable representations are as follow:

$X_{i,j}(2 \leq i \leq j \leq n)$:

$$\begin{array}{c}
 0 \\
 \searrow \\
 0 \longrightarrow \cdots \longrightarrow 0 \xrightarrow{i} k \xrightarrow{1} k \longrightarrow \cdots \xrightarrow{j} k \longrightarrow 0 \longrightarrow \cdots \longrightarrow 0 \\
 \nearrow \\
 0
 \end{array}$$

$Y_j(1 \leq j \leq n)$:

$$\begin{array}{c}
 k \\
 \searrow \\
 k \xrightarrow{1} \cdots \xrightarrow{j} k \longrightarrow 0 \longrightarrow 0 \longrightarrow \cdots \longrightarrow 0 \\
 \nearrow \\
 0
 \end{array}$$

3.4 Minimal Projective Resolution for Type D_n

$Y'_j (1 \leq j \leq n) :$

$$\begin{array}{c} 0 \\ \searrow \\ k \xrightarrow{1} \cdots \xrightarrow{1} k \rightarrow 0 \rightarrow 0 \rightarrow \cdots \rightarrow 0 \\ \nearrow \\ k \end{array}$$

j

$Z_j (2 \leq j \leq n) :$

$$\begin{array}{c} k \\ \searrow \\ k \xrightarrow{1} \cdots \xrightarrow{1} k \rightarrow 0 \rightarrow 0 \rightarrow \cdots \rightarrow 0 \\ \nearrow \\ k \end{array}$$

j

$T_{j,l} (2 \leq j < l \leq n) :$

$$\begin{array}{c} k \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \\ \searrow \\ k^2 \xrightarrow{I_2} \cdots \xrightarrow{I_2} k^2 \xrightarrow{(1 \ 1)} k \xrightarrow{1} k \rightarrow \cdots \xrightarrow{1} k \rightarrow 0 \rightarrow \cdots \rightarrow 0 \\ \nearrow \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \\ k \end{array}$$

$j \qquad l$

3.4 Minimal Projective Resolution for Type D_n

The following theorem comes from calculation:

Theorem 3.4.1 The minimal projective resolution of the previous indecomposable representations are as follow:

$$0 \rightarrow P(j+1) \rightarrow P(i) \rightarrow X_{i,j} \rightarrow 0$$

$$0 \rightarrow P(j+1) \rightarrow P(1) \rightarrow Y_j \rightarrow 0$$

$$0 \rightarrow P(j+1) \rightarrow P(1') \rightarrow Y'_j \rightarrow 0$$

$$0 \rightarrow P(2) \oplus P(j+1) \rightarrow P(1) \oplus P(1') \rightarrow Z_j \rightarrow 0$$

$$0 \rightarrow P(j) \oplus P(l+1) \rightarrow P(1) \oplus P(1') \rightarrow T_{j,l} \rightarrow 0$$

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4. Conclusion

What we've done:

- Investigate the basic theory of quiver representation theory.
- Calculate the minimal projective resolution of indecomposable representations of quivers of type A_n and a special class of type D_n .

What can be done in the future:

- Calculate the minimal projective resolution for all indecomposable representations.

We haven't done this since the computing all indecomposable representations for all finite type quivers is quite difficult. As an illustration, calculating all indecomposable representations for E_8 using AR-translations requires 120 steps[3].

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